

Exercise: 10.2

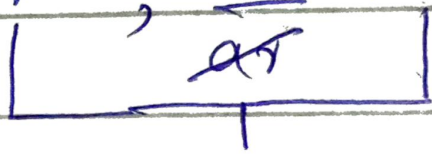
Finding the n th partial sums.

Geometric series:

$$a + ar + ar^2 + \dots + ar^{n-1} + \dots$$

if two consecutive no.s give same answer ^{by division,} then it is called geometric series.

e.g. $\frac{ar}{a} = r$, $\frac{ar^2}{ar} = r$

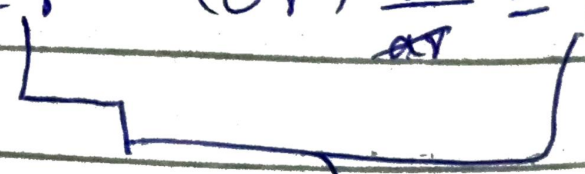


same answer.

Convergence of geometric series:

If the answer of series is between $(-1$ to $+1)$ then, the series converges.

for-example: $\frac{ar}{a} = r$ (or) $\frac{ar^2}{ar} = r$



$(-1 < r < 1) \rightarrow \text{converge}$

Divergence of geo-series:

if the answer (r) is b/w $(-\infty \text{ to } -1)$ and $(1 \text{ to } \infty)$ then the series diverges. $(|r| \geq 1)$

$$(-\infty, -1] \cup [1, \infty) \quad \downarrow \text{diverge}$$

$a \rightarrow$ first term of series.

$(1-r) \rightarrow$ common ratio

Nth partial sums: $\longleftrightarrow$

Question No. (1-6)

$$1) \quad 2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \dots + \frac{2}{3^{n-1}} + \dots$$

Sol) $a = 2$

$$\frac{\frac{2}{3}}{\frac{2}{9}} = \frac{1}{3}, \quad \frac{2}{9} \times \frac{3}{2} = \frac{1}{3}$$

So, $r = \frac{1}{3}$ ~~is~~ $(|r| < 1)$ so it converges.

$$\therefore S_n = \frac{a(1-r^n)}{(1-r)}$$

$$S_n = \frac{2(1 - (\frac{1}{3})^n)}{1 - (\frac{1}{3})}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{2(1 - (\frac{1}{3})^n)}{1 - \frac{1}{3}}$$

$$= \frac{2(1 - 0)}{\frac{2}{3}}$$

$$= 3$$

$$2) \frac{9}{100} + \frac{9}{100^2} + \frac{9}{100^3} + \dots + \frac{9}{100^n} + \dots$$

$$a = \frac{9}{100}, \quad r = \frac{\frac{9}{100}}{\frac{9}{100}}$$

So, $r = \frac{1}{100}$ ($-1 < r < 1$) so it converges to $\left(\frac{a}{1-r}\right)$.

$$S_n = \frac{a(1-r^n)}{(1-r)}$$

$$\lim_{n \rightarrow \infty} S_n = \frac{\frac{9}{100} \left(1 - \left(\frac{1}{100} \right)^n \right)}{1 - \frac{1}{100}}$$

$$= \frac{\frac{9}{100}}{\frac{99}{100}} = \frac{1}{11}$$

$$S_n = \frac{1}{11}$$

$$3) 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots + (-1)^{n-1} \frac{1}{2^{n-1}} + \dots$$

$$a = 1, r = \frac{-1/2}{1} = -\frac{1}{2}$$

$$(or) r = \frac{1/4}{-1/2} = \boxed{-\frac{1}{2} < 1}$$

So, the series convergent.

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$\lim_{n \rightarrow \infty} S_n = \frac{1 \left(1 - \left(-\frac{1}{2} \right)^n \right)}{1 + \frac{1}{2}}$$

$$S_n = \frac{1}{3/2}$$

$$S_n = \frac{2}{3}$$

✓

4) $1 - 2 + 4 - 8 + \dots + (-1)^{n-1} 2^{n-1} + \dots$

sol) $a=1$, $r = -2$ diverges.

~~$|r| < 1$~~ so the ~~convergent~~

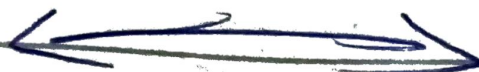
$(|r| < 1) S_n = \frac{a(1-r^n)}{1-r}$

$$\lim_{n \rightarrow \infty} S_n = \frac{1(1-(-2)^n)}{1-(-2)}$$

$$= \frac{1(1-0)}{1+2}$$

$$S_n = \frac{1}{3}$$

✓



$$5) \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \dots + \frac{1}{(n+1)(n+2)} + \dots$$

Sol) we use partial fraction in this type of series.

$$\frac{1}{(n+1)(n+2)} = \frac{A}{n+1} + \frac{B}{n+2}$$

$$= A(n+2) + B(n+1) \text{ --- (i)}$$

$$\text{put } n+1 = 0$$

($n = -1$) in eqn i.

$$= A(-1+2) + B(-1+1)$$

$$\boxed{1 = A}$$

Then put $n+2 = 0$

$$\boxed{n = -2}$$

$$1 = A(-2+2) + B(-2+1)$$

$$1 = B(-1)$$

$$\boxed{B = -1}$$

$$\frac{1}{(n+1)(n+2)} = \frac{1}{n+1} + \left(\frac{-1}{n+2} \right)$$

$$\frac{1}{(n+1)(n+2)} = \frac{1}{n+1} - \frac{1}{n+2}$$

so, according to question first
put $n=1$ then $n=2$ and so, on

$$= \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right)$$

$$\dots \left(\frac{1}{n+1} - \frac{1}{n+2} \right) \dots$$

$$= \frac{1}{2} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{4}} + \cancel{\frac{1}{4}} - \cancel{\frac{1}{5}} + \dots \left(\frac{1}{n+1} - \frac{1}{n+2} \right) \dots$$

$$S_n = \frac{1}{2} - \frac{1}{n+2}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(\frac{1}{2} - \frac{1}{n+2} \right)$$

$$= \frac{1}{2} - \frac{1}{\infty+2}$$

$$= \frac{1}{2} - \frac{1}{\infty} = \frac{1}{2} - 0 = \boxed{\frac{1}{2}}$$

(converges)

$$6) \frac{5}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \frac{5}{3 \cdot 4} + \dots + \frac{5}{n(n+1)} + \dots$$

we use partial fraction.

$$\frac{5}{n(n+1)} = \frac{A}{n} + \frac{B}{n+1}$$

$$5 = A(n+1) + B(n)$$

put $n=0$

$$5 = A(0+1) + B(0)$$

$$\boxed{A = 5}$$

Then put $n+1=0$; $n=-1$

$$5 = A(-1+1) + B(-1)$$

$$5 = -B$$

$$\boxed{B = -5}$$

$$\frac{B}{n(n+1)} = \frac{5}{n} - \frac{5}{n+1}$$

put $n=1$ then $n=2$, so on...

$$= \left(5 - \frac{5}{2}\right) + \left(\frac{5}{2} - \frac{5}{3}\right) + \left(\frac{5}{3} - \frac{5}{4}\right) \dots \left(\frac{5}{n} - \frac{5}{n+1}\right)$$

$$S_n = 5 - \frac{5}{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(5 - \frac{5}{n+1} \right)$$

$$= 5 - \frac{5}{\infty + 1} = 5 - \frac{5}{\infty}$$

$$S_n = 5 - 0 = \boxed{5}$$

(5) is a finite no. so, (convergent) A.

Question No. (15-18)

$$15) 1 + \left(\frac{2}{5}\right) + \left(\frac{2}{5}\right)^2 + \left(\frac{2}{5}\right)^3 + \left(\frac{2}{5}\right)^4 + \dots$$

$$\text{sol) } a = 1, r = \frac{2/5}{1} = \frac{2}{5}$$

$|r| > 1$ so, the series is divergent. b

$$16) 1 + (-3) + (-3)^2 + (-3)^3 + (-3)^4 + \dots$$

$$a = 1, r = -3$$

$$(-\infty, -1] \cup [1, \infty)$$

$|r| > 1$ so, the series is divergent.

$$17) \left(\frac{1}{8}\right) + \left(\frac{1}{8}\right)^2 + \left(\frac{1}{8}\right)^3 + \left(\frac{1}{8}\right)^4 + \dots$$

$$a = \frac{1}{8}, r = \frac{\frac{1}{64}8}{\frac{1}{8}}$$

$$\boxed{r = \frac{1}{8}}$$

since $|r| < 1$ so, the series is convergent.

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$\lim_{n \rightarrow \infty} S_n = \frac{\frac{1}{8} \left(1 - \left(\frac{1}{8}\right)^n\right)}{1 - \frac{1}{8}}$$

$$= \frac{1}{8} \left(\frac{1-0}{7/8} \right)$$

$$S_n = \frac{1}{7}$$

Ans.

$$18) \left(-\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^3 + \left(-\frac{2}{3}\right)^4 + \dots$$

$$a = \left(-\frac{2}{3}\right)^2 = \frac{4}{9}$$

$$r = \frac{\left(-\frac{2}{3}\right)^3}{\left(-\frac{2}{3}\right)^2} = \frac{-\frac{8}{27}}{\frac{4}{9}} = -\frac{2}{3}$$

$$r = -\frac{2}{3}$$

$|r| < 1$ so,
(convergent)

$$S_n = a(1-r^n)$$

$$1-r$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(\frac{\frac{4}{9} \left(1 - \left(-\frac{2}{3}\right)^n \right)}{1 + \frac{2}{3}} \right)$$

$$= \frac{4/9 \left(1 - \left(-\frac{2}{3}\right)^\infty \right)}{5/3}$$

$$= \frac{4/9 (1-0)}{5/3}$$

$$= \frac{4}{9} \times \frac{3}{5}$$

$$S_n = \frac{4}{15}$$

←————→
Telescoping Series

Q : (35 - 40)

Telescoping:

$$\therefore \sum_{n=1}^k \frac{1}{n(n+1)} = \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

(After partial fraction)

Q 35)

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

Sol) $S_n = \sum_{n=1}^n \left(\frac{1}{n} - \frac{1}{n+1} \right)$

put $n=1, n=2, \dots$

$$\Rightarrow \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$= 1 - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{4}} + \dots + \cancel{\frac{1}{n}} - \frac{1}{n+1}$$

$$S_n = 1 - \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right)$$

$$= 1 - \frac{1}{\infty+1} = 1 - \frac{1}{\infty}$$

$$= \boxed{1} \quad \text{(convergent)}$$

$$Q36) \sum_{n=1}^{\infty} \left(\frac{3}{n^2} - \frac{3}{(n+1)^2} \right)$$

$$\text{Sol)} S_n = \sum_{n=1}^n \left(\frac{3}{n^2} - \frac{3}{(n+1)^2} \right)$$

put $n=1, n=2, n=3, \dots$ so on...

$$= \left(\frac{3}{1} - \frac{3}{(2)^2} \right) + \left(\frac{3}{(2)^2} - \frac{3}{(3)^2} \right) + \dots + \left(\frac{3}{n^2} - \frac{3}{(n+1)^2} \right)$$

$$= 3 - \cancel{\frac{3}{4}} + \cancel{\frac{3}{4}} - \cancel{\frac{3}{9}} + \dots + \cancel{\frac{3}{n^2}} - \frac{3}{(n+1)^2}$$

$$S_n = 3 - \frac{3}{(n+1)^2}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(3 - \frac{3}{(n+1)^2} \right)$$

$$= 3 - \frac{3}{(\infty+1)^2}$$

$$= 3 - \frac{3}{(\infty)^2} \Rightarrow 3 - 0$$

$$= \boxed{3}$$

converges with

sum

$$\boxed{3}$$

✓

Q: 37) $\sum_{n=1}^{\infty} (\ln \sqrt{n+1} - \ln \sqrt{n})$
 sol) $S_n = \sum_{n=1}^n (\ln \sqrt{n+1} - \ln \sqrt{n})$

By expanding, $n=1, n=2, \dots$
 $= (\ln \sqrt{2} - \ln \sqrt{1}) + (\ln \sqrt{3} - \ln \sqrt{2}) + \dots$
 $(\ln \sqrt{n+1} - \ln \sqrt{n})$
 $= \ln \sqrt{2} - \ln \sqrt{1} + \ln \sqrt{3} - \ln \sqrt{2} + \dots$
 $(\ln \sqrt{n+1} - \ln \sqrt{n})$

$$S_n = \ln \sqrt{n+1} - \ln \sqrt{1}$$

$$S_n = \ln \sqrt{n+1} - 0$$

$$S_n = \ln \sqrt{n+1}$$

$$S_n = \ln (n+1)^{1/2}$$

$$S_n = \frac{1}{2} \ln (n+1)$$

$$\lim_{n \rightarrow \infty} S_n = \frac{1}{2} \cdot \lim_{n \rightarrow \infty} \ln (n+1)$$

$$= \frac{1}{2} \ln (\infty + 1)$$

$$S_n = \frac{1}{2} \ln (\infty) = \boxed{\infty}$$

(Diverges)

$$Q: 38) \sum_{n=1}^{\infty} (\tan(n) - \tan(n-1))$$

$$S_n = \sum_{n=1}^n (\tan(n) - \tan(n-1))$$

By expanding and putting $n=1, n=2, \dots$

$$= (\tan(1) - \tan(0)) + (\tan(2) - \tan(1)) + \dots + (\tan(n) - \tan(n-1))$$
$$= \cancel{\tan(1)} - \tan(0) + \cancel{\tan(2)} - \cancel{\tan(1)} + \dots + \tan(n) - \cancel{\tan(n-1)}$$

$$S_n = \tan(n) - \tan(0)$$

$$S_n = \tan(n)$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \tan(n)$$

$$= \tan(\infty)$$

Does not exist so
the series (diverges.)



$$Q:39) \sum_{n=1}^{\infty} \left(\cos^{-1} \left(\frac{1}{n+1} \right) - \cos^{-1} \left(\frac{1}{n+2} \right) \right)$$

$$\text{sol)} \quad S_n = \sum_{n=1}^n \left(\cos^{-1} \left(\frac{1}{n+1} \right) - \cos^{-1} \left(\frac{1}{n+2} \right) \right)$$

By expanding :-

$$= \left(\cos^{-1} \left(\frac{1}{2} \right) - \cos^{-1} \left(\frac{1}{3} \right) \right) + \left(\cos^{-1} \left(\frac{1}{3} \right) - \cos^{-1} \left(\frac{1}{4} \right) \right) \dots$$

$$\left(\cos^{-1} \left(\frac{1}{n+1} \right) - \cos^{-1} \left(\frac{1}{n+2} \right) \right)$$

$$= \cos^{-1} \left(\frac{1}{2} \right) - \cancel{\cos^{-1} \left(\frac{1}{3} \right)} + \cancel{\cos^{-1} \left(\frac{1}{3} \right)} - \cancel{\cos^{-1} \left(\frac{1}{4} \right)} \dots$$

$$\cancel{\cos^{-1} \left(\frac{1}{n+1} \right)} - \cancel{\cos^{-1} \left(\frac{1}{n+2} \right)}$$

$$\therefore S_n = \cos^{-1} \left(\frac{1}{2} \right) - \cos^{-1} \left(\frac{1}{n+2} \right)$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(\cos^{-1} \left(\frac{1}{2} \right) - \cos^{-1} \left(\frac{1}{n+2} \right) \right)$$

$$= \cos^{-1} \left(\frac{1}{2} \right) - \cos^{-1} \left(\frac{1}{\infty+2} \right)$$

$$= \cos^{-1} \left(\frac{1}{2} \right) - \cos^{-1} (0)$$

$$= \frac{\pi}{3} - \frac{\pi}{2} \Rightarrow \frac{2\pi - 3\pi}{6}$$

$$S_n = \left[-\frac{\pi}{6} \right] \text{ convergent } \Delta$$

Q: 40) $\sum_{n=1}^{\infty} (\sqrt{n+4} - \sqrt{n+3})$

Sol) $S_n = \sum_{n=1}^n (\sqrt{n+4} - \sqrt{n+3})$

By expanding :

$$= (\sqrt{5} - \sqrt{4}) + (\sqrt{6} - \sqrt{5}) + (\sqrt{7} - \sqrt{6}) + \dots$$

$$= \sqrt{5} - \sqrt{4} + \sqrt{6} - \sqrt{5} + \sqrt{7} - \sqrt{6} + \dots \sqrt{n+4} - \sqrt{n+3}$$

$$S_n = \sqrt{n+4} - \sqrt{4}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} (\sqrt{n+4} - \sqrt{4})$$

$$= \sqrt{\infty + 4} - \sqrt{4}$$

$$= \sqrt{\infty} - \sqrt{4}$$

$$= \infty - 2 = \boxed{\infty}$$

~~$S_n = 2$ (converges)~~

(Diverges)



Repeating decimals

Q: (19 — 26)

$$19) 0.\overline{23} = 0.23\ 23\ 23 \dots$$

$$= 0.23 + 0.0023 + 0.000023 + \dots$$

$$= 0 + \frac{23}{100} + \frac{23}{(100)^2} + \frac{23}{(100)^3} + \dots$$

$$= 0 + \frac{23}{100} \left(1 + \frac{23}{100} + \frac{23}{(100)^2} + \frac{23}{(100)^3} + \dots \right)$$

$$\boxed{a=1}, \quad r = \frac{\frac{23}{(100)^2}}{\frac{23}{100}} = \frac{1}{100}$$

$$\boxed{r=0.01}$$

$$S_n = \frac{a}{1-r} = \frac{23}{1-0.01}$$

$$= 0 + \frac{23}{100} \left(\frac{1}{1-0.01} \right)$$

$$= \frac{23}{100} \left(\frac{1}{0.99} \right) \Rightarrow \frac{23}{99}$$

$$= 0.\overline{23} \rightarrow \boxed{\begin{array}{r} 23 \\ 99 \end{array}} \quad \checkmark$$

$$21) \quad 0.\overline{7} = 0.7 + 0.07 + 0.007 + \dots$$

$$= 0 + \frac{7}{10} + \frac{7}{(10)^2} + \frac{7}{(10)^3} + \dots$$

$$= 0 + \frac{7}{10} \left(1 + \frac{7}{(10)} + \frac{7}{(10)^2} + \dots \right)$$

$$a=1, \quad r = \frac{\frac{7}{(10)^2}}{\frac{7}{10}} = \frac{1}{10}$$

$$\boxed{r = 0.1}$$

$$S_n = \frac{a}{1-r} = \frac{1}{1-0.1}$$

$$= 0 + \frac{7}{10} \left(\frac{1}{1-0.1} \right)$$

$$= \frac{7}{10} \left(\frac{1}{0.9} \right) \Rightarrow \boxed{\frac{7}{9}} \quad \text{b}$$

$$\text{So, } 0.\overline{7} \rightarrow \frac{7}{9} \quad \text{d}$$



$$\begin{aligned}
 25) \quad 1.24\overline{123} &= 1.24\ 123\ 123\ 123 \dots \\
 &= 1.24 + 0.00123 + 0.00000123 + \dots \\
 &= \cancel{1.24} + \frac{123}{100000} + \frac{123}{100000000} + \dots
 \end{aligned}$$

$$= 1.24 + \frac{123}{100000} + \frac{123}{(100000)^2} + \frac{123}{(100000)^3} + \dots$$

$$= 1.24 + \frac{123}{100000} \left(1 + \frac{123}{100000} + \frac{123}{(100000)^2} + \dots \right)$$

$$\cancel{1.24} \quad a=1, \quad r = \frac{\frac{123}{100000}}{\frac{123}{100000}} = r = \frac{1}{100000}$$

$$\boxed{r = 0.00001}$$

$$S_n = \frac{a}{1-r} = \frac{1}{1-0.00001}$$

$$= 1.24 + \frac{123}{100000} \left(\frac{1}{1-0.00001} \right)$$

$$= 1.24 + \frac{123}{99999}$$

$$= \frac{124}{100} + \frac{123}{99999}$$

$$= \frac{124 \times 99999 + 123 \times 100}{9999900}$$

$$= \frac{12412176}{9999900} = 1.24123 \dots$$

So, $1.24\overline{123} \rightarrow$

nTh term divergence Test

(Ex 10.2) @ (27-34)

Take ~~$\lim_{n \rightarrow \infty} a_n = 0$~~ $\Rightarrow$ (convergent)

$\lim_{n \rightarrow \infty} a_n \neq 0 \rightarrow$ (divergent)

Exam:

$$\sum_{n=1}^{\infty} n^2$$

$$\lim_{n \rightarrow \infty} (\infty)^2 = \infty \text{ diverge.}$$

(nth term divergence) Test

Q: (27 — 34)

$$27) \sum_{n=1}^{\infty} \frac{n}{n+10} \Rightarrow a_n = \frac{n}{n+10}$$

$$\text{sol } a_n = \lim_{n \rightarrow \infty} \left(\frac{n}{n+10} \right)$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{n(1 + \frac{10}{n})} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{10}{n}}$$

$$= \frac{1}{1 + \frac{10}{\infty}} = \frac{1}{1 + 0} = 1 \neq 0 \quad (\text{divergent})$$

$$28) \sum_{n=1}^{\infty} \frac{n(n+1)}{(n+2)(n+3)}$$

$$a_n = \frac{n(n+1)}{(n+2)(n+3)}$$

$\left(\frac{\infty}{\infty} \right)$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{n^2 + n}{n^2 + 5n + 6} \right)$$

$\hookrightarrow$ The maximum power

is n^2 so, divide by n^2 .

(or) use L'Hopital rule.

$$\lim_{n \rightarrow \infty} \left(\frac{2n+1}{2n+5} \right)$$

$$\lim_{n \rightarrow \infty} \frac{2n \left(1 + \frac{1}{2n} \right)}{2n \left(1 + \frac{5}{2n} \right)}$$

$$= \frac{1 + \frac{1}{\infty}}{1 + \frac{5}{\infty}} \Rightarrow \frac{1+0}{1+0}$$

$$= 1 \neq 0 \text{ so, (divergent)}$$

$$29) \sum_{n=0}^{\infty} \frac{1}{n+4}$$

$$a_n = \frac{1}{n+4}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n+4}$$

$$= \frac{1}{\infty+4} \Rightarrow \frac{1}{\infty}$$

$$a_n = 0 \text{ (Not diverge). } \checkmark$$

Also, ~~does~~ this test does n't prove the series converges. To determine convergence, you need to use such as, integral test, comparison etc.

$$30) \sum_{n=1}^{\infty} \frac{n}{n^2+3}$$

$$a_n = \frac{n}{n^2+3}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{n}{n^2+3} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{n}{n(n+3/n)}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n+3/n}$$

$$= \frac{1}{\infty + 3/\infty} \Rightarrow \frac{1}{\infty + 0}$$

$$= \frac{1}{\infty} = 0 \quad \text{Not diverge}$$

$$31) \sum_{n=1}^{\infty} \frac{\cos \frac{1}{n}}{n}$$

$$a_n = \frac{\cos \frac{1}{n}}{n}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \cos\left(\frac{1}{n}\right)$$

$$= \cos\left(\frac{1}{\infty}\right) \Rightarrow \cos(0)$$

$$= 1 \neq 0 \quad \text{Diverges}$$

$$32) \sum_{n=0}^{\infty} \frac{e^n}{e^n + n}$$

$$a_n = \frac{e^n}{e^n + n}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{e^n}{e^n + n} \quad \left(\frac{\infty}{\infty} \right)$$

~~l'Hopital~~ l'Hopital Rule

$$= \lim_{n \rightarrow \infty} \frac{e^n}{e^n + 1} \rightarrow \text{again } \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{e^n}{e^n} \right) \Rightarrow \lim_{n \rightarrow \infty} (1)$$

$$= 1 \neq 0 \text{ so, } a_n \rightarrow 1 \text{ (diverge) } \Delta$$

$$33) \sum_{n=1}^{\infty} \ln \frac{1}{n}$$

$$\text{Sol)} a_n = \ln \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \ln \frac{1}{n}$$

$$= \ln \left(\frac{1}{\infty} \right) \Rightarrow \ln(0)$$

$$= \ln(0) = -\infty \Delta$$

~~##~~ (Diverges)

Convergence or Divergence

Ex: 10.2 (49 — 68)

$$49) \sum_{n=0}^{\infty} \left(\frac{1}{\sqrt{2}} \right)^n$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{2}} \right)^n$$

$$= \sum_{n=0}^{\infty} 1 \left(\frac{1}{\sqrt{2}} \right)^n$$

$$|r| = 0.707 < 1$$

So, convergent.

$$S_n = \frac{a}{1-r}$$

$$S_n = \frac{1}{1-0.707}$$

$$S_n = 3.41$$

So, the sequence converges to 3.41, because $|r| < 1$.

$$50) \sum_{n=0}^{\infty} (\sqrt{2})^n$$

$$= \lim_{n \rightarrow \infty} (\sqrt{2})^n$$

$$= \sum_{n=0}^{\infty} 1(\sqrt{2})^n$$

$$a=1, r=\sqrt{2} > 1$$

$|r| > 1$ so, diverges to $\frac{a}{1-r}$

$$S_n = \frac{a}{1-r} = \frac{1}{1-\sqrt{2}}$$

$$\frac{a}{1-r} = \boxed{S_n = -2.43}$$

✓

$$51) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{3}{2^n}$$

$$= \lim_{n \rightarrow \infty} (-1)^{n+1} \frac{3}{2^n}$$

$$= \lim_{n \rightarrow \infty} (-1)^n (-1) \frac{3}{2^n}$$

$$= \lim_{n \rightarrow \infty} (-1)^n \left(-\frac{3}{2^n} \right)$$

$$= \lim_{n \rightarrow \infty} -3 \left(\frac{-1}{2} \right)^n$$

$$= \sum_{n=1}^{\infty} -3 \left(\frac{-1}{2} \right)^n$$

$r = -\frac{1}{2}$; $|r| < 1$, so it converges to $\frac{a}{1-r}$.

To find a , put $(n=1)$.

$$= \sum_{n=1}^{\infty} -3 \left(-\frac{1}{2} \right)^n$$

$$= \textcircled{-3} \left(-\frac{1}{2} \right)^1 = 3/2$$

So, $a = \frac{3}{2}$

$$\frac{a}{1-r} = \frac{3/2}{1+(\frac{1}{2})}$$

$$= \frac{3/2}{3/2} = 1$$

So, $|r| < 1$ and it converges to $\frac{a}{1-r} = 1$.

52) $\sum_{n=1}^{\infty} (-1)^{n+1} n$

~~This~~ This is not a geometric series so, Here we apply nth term divergence test.

$$a_n = (-1)^{n+1} n$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (-1)^{n+1} n$$

$$= (-1)^{\infty+1} (\infty)$$

$$= (\infty) \text{ or } (-\infty)$$

$a_n \neq 0$ so, the series diverges.

53) $\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$

This series is not a geometric and n th term test cannot apply on it.

So, put $n=0$.

$$\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$$

$$= \cos(0)$$

$= 1$, then put $n=2, n=3, \dots$

$$= \cos\left(\frac{2\pi}{2}\right) = 0$$

$$= \cos\left(\frac{3\pi}{2}\right) = -1$$

So, the series is $1+0-1+0\dots$

The series diverges.



$$54) \sum_{n=0}^{\infty} \frac{\cos n\pi}{5^n}$$

if we put $n=1, n=2, n=3, \dots$
 $\cos$ gives ~~even~~^{+ve} answer if n is even and if n is odd $\cos$ gives -ve answer. So,

$$1 - 1 + 1 - 1 + 1 - 1 \dots = (-1)^n$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{5^n}$$

$$= \sum_{n=0}^{\infty} 1 \left(-\frac{1}{5}\right)^n$$

$$r = -\frac{1}{5}, |r| < 1, \text{ so}$$

the series converges to $\frac{a}{1-r}$.

$$S_n = \frac{a}{1-r} = \frac{1}{1 + \frac{1}{5}} \Rightarrow \frac{5}{6}$$

$$55) \sum_{n=0}^{\infty} e^{-2n}$$

$$= \sum_{n=0}^{\infty} (e^{-2})^n$$

$$r = e^{-2} = \frac{1}{e^2}$$

$|r| < 1$ so, the series converges to $\frac{a}{1-r}$.

~~to find~~ $\frac{a}{1-r} = \frac{1}{1 - \frac{1}{e^2}}$

$$S_n = \frac{e^2}{e^2 - 1} \quad \text{Ans.}$$

$$56) \sum_{n=1}^{\infty} \ln \frac{1}{3^n}$$

$$= \sum_{n=1}^{\infty} (\ln(1) - \ln 3^n)$$

$$= \sum_{n=1}^{\infty} (-\ln 3^n)$$

$$= \sum_{n=1}^{\infty} -n \ln(3)$$

This is not a geometric series so, apply nth

term divergence test.

$$a_n = -n \ln(3)$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (-n \ln(3)) \\ = -\infty \ln(3)$$

$$\lim_{n \rightarrow \infty} a_n = -\infty$$

so, $a_n \neq 0$, the series diverges ϕ .

$$57) \sum_{n=1}^{\infty} \frac{2}{10^n}$$

$$= \sum_{n=1}^{\infty} 2 \left(\frac{1}{10} \right)^n$$

$$r = \frac{1}{10}; |r| < 1, \text{ so}$$

the series converges to $\frac{a}{1-r}$.

To find a , put $n=1$.

$$a = \frac{2}{10}$$

$$\frac{a}{1-r} = \frac{\frac{2}{10}}{1 - \frac{1}{10}} = \frac{2}{9} \phi$$

$$58) \sum_{n=0}^{\infty} \frac{1}{x^n} ; |x| > 1$$

$$= \sum_{n=0}^{\infty} 1 \left(\frac{1}{x} \right)^n$$

$$r = \frac{1}{x} ; \cancel{|x| > 1} \text{ so}$$

$|r| < 1$, when $|x| > 1$ so
the series converges to $\frac{a}{1-r}$

$$\frac{a}{1-r} = \frac{1}{1 - \frac{1}{x}} = \frac{x}{x-1}$$

$$59) \sum_{n=0}^{\infty} \frac{2^n - 1}{3^n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{2^n}{3^n} - \frac{1}{3^n} \right)$$

$$= \sum_{n=0}^{\infty} \left(\left(\frac{2}{3} \right)^n - \left(\frac{1}{3} \right)^n \right)$$

$$= \sum_{n=0}^{\infty} 1 \left(\frac{2}{3} \right)^n - \sum_{n=0}^{\infty} 1 \left(\frac{1}{3} \right)^n$$

$$r = \frac{2}{3} ; |r| < 1$$

$$r = \frac{1}{3} ; |r| < 1$$

$$= \frac{1}{1 - 2/3} - \frac{1}{1 - 1/3}$$

$$= 3 - \frac{3}{2} = \frac{3}{2}$$

So, the series converges to $\frac{3}{2}$.

$$60) \sum_{n=1}^{\infty} \left(1 - \frac{1}{n}\right)^n$$

$$= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n$$

$= e \neq 0$, so, the series diverges.

$$61) \sum_{n=0}^{\infty} \frac{n!}{1000^n}$$

$$\lim_{n \rightarrow \infty} \frac{n^n}{n!} = 0$$

$$\text{So, } \sum_{n=0}^{\infty} \frac{n!}{1000^n}$$

$$= \sum_{n=0}^{\infty} \frac{\frac{1}{1000^n}}{\frac{1}{n!}} \cdot \frac{1}{1000^n}$$

$$= \sum_{n=0}^{\infty} \frac{1}{1000^n} \cdot \frac{1}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{\left(\frac{1}{1000}\right)^n}{n!}$$

$$a_n = \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{1000}\right)^n}{n!}$$

~~$$= \frac{\left(\frac{1}{1000}\right)^{\infty}}{\infty} = 0$$~~

$$= \frac{\left(\frac{1}{1000}\right)^{\infty}}{\infty} = \infty$$

So, $a_n \neq 0$, the series diverges.

$$62) \sum_{n=1}^{\infty} \frac{n^n}{n!}$$

use n th term divergence test.

$$\begin{aligned} n^n &= n \cdot n \cdot n \cdot n \cdots n \text{ times} \\ &= \sum_{n=1}^{\infty} \frac{n \cdot n \cdot n \cdots n}{1 \cdot 2 \cdot 3 \cdot 4 \cdots n} \end{aligned}$$

$$\begin{aligned} a_n &= \lim_{n \rightarrow \infty} n^{n-1} \\ &= (\infty)^{\infty-1} = \infty \end{aligned}$$

So, the series diverges.

$$63) \sum_{n=1}^{\infty} \frac{2^n + 3^n}{4^n}$$

$$= \sum_{n=1}^{\infty} \frac{2^n}{4^n} + \sum_{n=1}^{\infty} \frac{3^n}{4^n}$$

$$= \sum_{n=1}^{\infty} \left(\frac{2}{4}\right)^n + \sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n$$

$$r = \frac{2}{4}; |r| < 1,$$

$$r = \frac{3}{4}; |r| < 1$$

= conv - conv

= convergent

$$64) \sum_{n=1}^{\infty} \frac{2^n + 4^n}{3^n + 4^n}$$

= Divide by 4^n

$$= \sum_{n=1}^{\infty} \frac{\frac{2^n}{4^n} + 1}{\frac{3^n}{4^n} + 1}$$

$$= \sum_{n=1}^{\infty} \frac{\left(\frac{2}{4}\right)^n + 1}{\left(\frac{3}{4}\right)^n + 1}$$

$$a_n = \frac{\left(\frac{2}{4}\right)^n + 1}{\left(\frac{3}{4}\right)^n + 1}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{\left(\frac{1}{2}\right)^n + 1}{\left(\frac{3}{4}\right)^n + 1} \right)$$

$$= \frac{\left(\frac{1}{2}\right)^{\infty} + 1}{\left(\frac{3}{4}\right)^{\infty} + 1} = \frac{0 + 1}{0 + 1}$$

= 1, so, $a_n \neq 0$, the Series diverges

$$65) \sum_{n=1}^{\infty} \ln \left(\frac{n}{n+1} \right)$$

$$S_n = \sum_{n=1}^{\infty} (\ln(n) - \ln(n+1))$$

use Telescopic series.

$$= (\ln(1) - \ln(2)) + (\ln(2) - \ln(3)) + \dots + (\ln(n) - \ln(n+1))$$

$$= \ln(1) - \ln(n+1)$$

$$S_n = -\ln(n+1)$$

$$= \lim_{n \rightarrow \infty} (-\ln(n+1))$$

$$= -\ln(\infty + 1) = \boxed{-\infty}$$

(Diverges.)

$$66) \sum_{n=1}^{\infty} \ln \left(\frac{n}{2n+1} \right)$$

$$= \sum_{n=1}^{\infty} \ln(n) - \ln(2n+1)$$

$$= \ln(1) - \ln(2(1)+1)$$

$$= 0 - \ln(3)$$

$$= -\ln(3) \neq 0$$

So, the series diverges.

$$67) \sum_{n=0}^{\infty} \left(\frac{e}{\pi} \right)^n$$

$$r = \frac{e}{\pi} ; |r| < 1$$

So, the series converges to $\frac{a}{1-r}$.

$$\frac{a}{1-r} = \frac{1}{1 - \frac{e}{\pi}} = \frac{\pi}{\pi - e}$$



$$68) \sum_{n=0}^{\infty} \frac{e^{n\pi}}{\pi^{ne}}$$

$$= \sum_{n=0}^{\infty} \left(\frac{e^{\pi}}{\pi^e} \right)^n$$

$r = \frac{e^{\pi}}{\pi^e}$; ~~if~~ if we calculate
common ratio then

$|r| > 1$ so, the series
diverges.

